Analysis of Robustness of Linear Estimators via ℓ^p and Dual Norm Lipschitz Bounds

Clement (Kai Hei) Chu*
University of Toronto

May 22, 2025

Abstract

The sample mean is a fundamental estimator in statistical science, particularly known for its unbiasedness in its estimation of the population mean μ. While robustness is often studied through measures such as mean squared error and consistency, this paper adopts a different lens of analyzing robustness through Lipschitz continuity under various ℓ^p norms. We investigate how small perturbations in data affect linear estimators by bounding their sensitivity via dual norms. In particular, we show that the sample mean, as a linear functional, exhibits Lipschitz stability, and we extend this analysis to general linear estimators. Further, we characterize symmetric linear estimators and explore generalizations involving dual norms.

*Email: `clement.chu@mail.utoronto.ca`

Acknowledgements

I would like to thank Sin Hang Jason Yeung (PhD candidate, University of Toronto, Department of Mathematics) for generously taking the time to give incredibly valuable comments and advice for previous versions of this paper.

1 Preliminary Definitions, and Key Inequalities

Definition 1.1 (ℓ^p-norm [1]). *Fix $p \in \{1, \dots, \infty\}$. The ℓ^p-type-norm on $\mathbb{R}^n$ is defined as*

$$\|x\|_p := \begin{cases} \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}, & \text{for } 1 \leq p < \infty, \\ \max_{1 \leq i \leq n} |x_i|, & \text{for } p = \infty. \end{cases}$$

Notation 1.2. *Throughout the entire paper, we would equip $\mathbb{R}^n$ with the ℓ^p-type-norm as defined in Definition 1.1, and the normed space would be denoted as $(\mathbb{R}^n, \|\cdot\|_p)$.*

Definition 1.3 (Lipschitz continuity [6]). *Let $(X, d_1), (Y, d_2)$ be two metric spaces. Then, the function $f : X \to Y$ is defined as being K-Lipschitz continuous if there exist a real constant $K \geq 0$ such that for all $x_1, x_2 \in X$,*

$$d_2(f(x_1), f(x_2)) \leq K d_1(x_1, x_2)$$

Lemma 1.4 (Hölder inequality [7]). [1] *Let $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then for any vectors $x, v \in \mathbb{R}^n$, we have*

$$\left| \sum_{i=1}^n x_i v_i \right| \leq \|x\|_p \cdot \|v\|_q.$$

Notation 1.5. *For this paper, we use $x \in \mathbb{R}^n$ to denote a vector, and do not use boldface notation* $\mathbf{x}$ *unless explicitly stated.*

2 Introduction

The reader may question the use of Lipschitz continuity to analyze the quality of an estimator. While some factors that assess the quality of estimators are more than sufficient in most cases (for example, unbiasedness means that the estimator would estimate the correct quantity in the long run), in often fails to address the issue to measure how robust an estimator is against change in data inputs (thus, data perturbations). Observe that Lipschitz continuity is perfectly suited for this purpose since it quantifies how much "allowance" the estimator has for such errors. We shall see the first simple example of the sample mean.

3 Lipschitz Continuity of the Sample Mean

Theorem 3.1. *Let $x, y \in \mathbb{R}^n$. Let $\hat{\mu} : (\mathbb{R}^n, \|\cdot\|_p) \to (\mathbb{R}, |\cdot|)$ be defined as $\hat{\mu}(x) = \frac{1}{n} \sum_{i=1}^n x_i$. Then for all $p \in [1, \infty]$, we have*

$$|\hat{\mu}(x) - \hat{\mu}(y)| \leq n^{-\frac{1}{p}} \|x - y\|_p.$$

That is, the sample mean $\hat{\mu}$ is $n^{-\frac{1}{p}}$-Lipschitz continuous.

Proof. Let $w = \left(\frac{1}{n}, \dots, \frac{1}{n}\right) \in \mathbb{R}^n$. Thus, we can define $\hat{\mu}(x)$ as $\langle w, x \rangle$.[2] Then by Hölder's inequality (1.4), we have

$$|\hat{\mu}(x) - \hat{\mu}(y)| = |\langle w, (x - y) \rangle| \leq \|w\|_q \|(x - y)\|_p = (\sum_{i=1}^n |\frac{1}{n}|^q)^{\frac{1}{q}} \|(x - y)\|_p.$$

[1] Here, we used $\mathbb{R}^n$ rather than a general measure space to maintain clarity.
[2] Notice that this matches the notion of $\hat{\mu}$ being a linear functional in $\mathbb{R}^n$.

Since $n > 0$,

$$(\sum_{i=1}^{n} |\frac{1}{n}|^q)^{\frac{1}{q}} \|(x-y)\|_p = (n^{1-p})^{\frac{1}{p}} \|(x-y)\|_p = n^{\frac{1}{q}-1} \|(x-y)\|_p$$

Since $\frac{1}{p} + \frac{1}{q} = 1$,

$$n^{\frac{1}{q}-1} \|(x-y)\|_p = n^{1-\frac{1}{p}-1} \|(x-y)\|_p = n^{-\frac{1}{p}} \|(x-y)\|_p$$

□

Why is this important? Notice that $\hat{\mu}$ being $n^{-\frac{1}{p}}$-Lipschitz continuous means that the **Lipschitz constant** (So here, the Lipschitz constant is $n^{-\frac{1}{p}}$) will decrease as n increases. More specifically, if we have an error in the data, which is depicted here by $\|(x-y)\|_p$, it would mean that as the sample size n gets larger, the error in the output would be much less than the errors in the input, thus very robust against data perturbations.

Remark 3.2. In fact, the lipschitz constant $n^{-\frac{1}{p}}$ is the smallest such bound on the inequality. In particular, notice that if $x = (1, \ldots, 1) \in \mathbb{R}^n$ and $y = (0, \ldots, 0) \in \mathbb{R}^n$, then $\hat{\mu}(x) = 1, \hat{\mu}(y) = 0$. Also, $\|x - y\|_p = \|(1, \ldots, 1) - (0, \ldots, 0)\|_p = \|(1, \ldots, 1)\|_p = (\sum_{i=1}^n 1^p)^{\frac{1}{p}} = n^{\frac{1}{p}}$. Hence, notice the equality is achieved, that is, $|\hat{\mu}(x) - \hat{\mu}(y)| = 1 = n^{-\frac{1}{p}}(n^{\frac{1}{p}}) = n^{-\frac{1}{p}}\|x - y\|_p$.

Also, Lipschitz offers a more concrete representation of continuity. In particular, Lipschitz continuity implies continuity. We first let the reader revisit the classic definition of continuity, then introduce the proof.

Definition 3.3 (Continuity). *A function $f : (X, d_1) \to (Y, d_2)$ is said to be continuous at a point $x_0 \in X$ if and only if*

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x \in X,\ d_1(x, x_0) < \delta \Rightarrow d_2(f(x), f(x_0)) < \varepsilon.$$

We will prove that Lipschitz continuity implies continuity now.

Theorem 3.4. *If $f : (X, d_1) \to (Y, d_2)$ is Lipschitz continuous, then f is continuous everywhere.*

Proof: Fix $\varepsilon > 0$. Since f is Lipschitz continuous on X, there exist a real constant $K \geq 0$ such that for all $x_1, x_2 \in X, d_2(f(x_1), f(x_2)) \leq K d_1(x_1, x_2)$. (1.3) Choose $\delta = \frac{\varepsilon}{K}$. Let $x_0 \in X$ be arbitrary. We aim to show that $\forall x \in X$, $|x_0 - x| < \delta \implies |f(x_0) - f(x)| < \varepsilon$.

Since f is Lipschitz continuous, $d_2(f(x_0), f(x)) \leq K d_1(x_0, x) < K\delta = K(\frac{\varepsilon}{K}) = \varepsilon$. Since $x \in X$ was chosen arbitrarily, f is continuous on X. □

In fact, the reader may have observed that Lipschitz continuity implies uniform continuity as well.

Corollary 3.5. *If $f : (X, d_1) \to (Y, d_2)$ is Lipschitz continuous, then f is uniform continuous.*

Proof: It follows immediately from Theorem 3.4 since δ does not depend on x. □

4 General Linear Estimators

In fact, the sample mean as shown above is a **linear estimator** (Or more generally, a *linear functional* on $\mathbb{R}^n$). Here is the precise definition tailored to this context.

Definition 4.1. *[4] An estimator $\hat{\theta} : (X \subseteq \mathbb{R}^n, \|\cdot\|_p) \to (\mathbb{R}, |\cdot|)$ is a linear estimator if and only if it can be written as $\hat{\theta} = \sum_{i=1}^n a_i x_i = a^T \cdot x$, where $a \in \mathbb{R}^n$ and $x \in X$.*

Proposition 4.2. *The sample mean is a linear estimator.*

Proof: It follows immediately from Definition 4.1, where $a = (\frac{1}{n}, \ldots, \frac{1}{n})$. □

Could we generalize this to all linear estimators? Are all linear estimators lipschitz continuous? The answer is yes!

Theorem 4.3. *Let $x, y \in \mathbb{R}^n$. Let $\hat{\theta} : (\mathbb{R}^n, \|\cdot\|_p) \to (\mathbb{R}, |\cdot|)$ be a linear estimator defined as $\sum_{i=1}^n a_i x_i = a \cdot x$, where $a, x \in \mathbb{R}^n$. Then for all $p \in [1, \infty]$, we have*

$$|\hat{\theta}(x) - \hat{\theta}(y)| \leq \|a\|_q \|(x - y)\|_p$$

That is, all linear estimators in $\mathbb{R}^n$ are $\|a\|_q$-Lipschitz continuous on $\mathbb{R}^n$.

Proof:

$$|\hat{\theta}(x) - \hat{\theta}(y)| = |\sum_{i=1}^n a_i x_i - \sum_{i=1}^n a_i y_i| = |\sum_{i=1}^n a_i (x_i - y_i)| = |\langle a, (x - y) \rangle|$$

By Hölder's inequality (1.4),

$$|\langle a, (x - y) \rangle| \leq \|a\|_q \|(x - y)\|_p.$$

□

It turns out, also, that the quantity $\|a\|_q$ that we've been touching on frequently has significance as well. It is, the dual norm of the linear functional $\langle a, \cdot\rangle$!

Definition 4.4 (Dual norm [5]). *Let $f : (\mathbb{R}^n, \|\cdot\|_p) \to (\mathbb{R}, |\cdot|)$ be a linear functional. Then the **dual norm** of f is defined as*

$$\|f\| := \sup_{\substack{x\in\mathbb{R}^n \\ \|x\|_p\leq 1}} |f(x)|$$

We shall prove that $\|a\|_q$ is the dual norm of a.

Proposition 4.5. *Let $\langle a, \cdot\rangle$ be a linear functional on $(\mathbb{R}^n, \|\cdot\|_p)$. Then, $\|a\|_q$ is the dual norm of $\langle a, \cdot\rangle$.*

Proof. By Hölder's inequality (1.4), for all $x \in \mathbb{R}^n$ with $\|x\|_p \leq 1$, we have

$$\begin{aligned} |\langle a, x\rangle| &\leq \|a\|_q \cdot \|x\|_p \\ &\leq \|a\|_q \quad (\text{since } \|x\|_p \leq 1) \end{aligned}$$

Thus,

$$\sup_{\|x\|_p\leq 1} |\langle a, x\rangle| \leq \|a\|_q. \tag{1}$$

We seek a vector $x \in \mathbb{R}^n$ such that (1) achieves an equality. Let $x \in \mathbb{R}^n$ be defined as

$$x = \left(\left(\frac{|a_1|}{\|a\|_q}\right)^{\frac{1}{p-1}}\operatorname{sgn}(a_1), \ldots, \left(\frac{|a_n|}{\|a\|_q}\right)^{\frac{1}{p-1}}\operatorname{sgn}(a_n)\right) \in \mathbb{R}^n$$

We claim that $|\langle a, x\rangle| = \|a\|_q$. First, notice that $\|x\|_p = 1$. Since, $\|x\|_p = (\sum_{i=1}^n x_i^p)^{\frac{1}{p}} = 1 \implies \sum_{i=1}^n x_i^p = 1$. But, notice, $\sum_{i=1}^n x_i^p = \sum_{i=1}^n((\frac{|a_i|}{\|a\|_q})^{\frac{1}{p-1}}\operatorname{sgn}(a_i))^p = \sum_{i=1}^n((\frac{|a_i|}{\|a\|_q})^{q-1}\operatorname{sgn}(a_i))^p$(since $\frac{1}{p} + \frac{1}{q} = 1$). Also, $p(q-1) = \frac{q}{q-1}(q-1) = q$, so $\sum_{i=1}^n((\frac{|a_i|}{\|a\|_q})^{q-1}\operatorname{sgn}(a_i))^p = \frac{\sum_{i=1}^n |a_i|^q}{\|a\|_q^q} = \frac{\|a\|_q^q}{\|a\|_q^q} = 1$.

We now show that $|\langle a, x\rangle| = \|a\|_q$. Note that, $a_i x_i = a_i((\frac{|a_i|}{\|a\|_q})^{\frac{1}{p-1}}\operatorname{sgn}(a_i)) = a_i((\frac{|a_i|}{\|a\|_q})^{q-1}\operatorname{sgn}(a_i)) = \frac{|a_i|^{q-1}|a_i|}{\|a\|_q^{q-1}}$(since $a_i(\operatorname{sgn}(a_i)) = |a_i|$) $= \frac{|a_i|^q}{\|a\|_q^{q-1}}$. Hence,

$$\sum_{i=1}^n a_i x_i = \sum_{i=1}^n \frac{|a_i|^q}{\|a\|_q^{q-1}} = \frac{\sum_{i=1}^n |a_i|^q}{\|a\|_q^{q-1}} = \frac{\|a\|_q^q}{\|a\|_q^{q-1}} = \|a\|_q$$

□

Why is this formulation using suprema useful? Notice that this represents the tighest bound on the linear functional. Since its the least upper bound by definition, the dual norm would therefore be the best, smallest Lipschitz constant that gives the tightest, and a very meaningful bound on the data perturbations.

5 Uniqueness of the Sample Mean

It is totally normal to question the limited examples we gave in this paper. While there are many useful linear estimators in statistics, there is a very specific reason why we chose to motivate the sample mean in the first place. In particular, we will prove that the sample mean minimizes the Lipschitz constant as given in Theorem 4.3, which is an effective way to demonstrate the robustness of the sample mean against data perturbations.

We will first prove that the $(\frac{1}{n}, \dots, \frac{1}{n})$ indeed minimizes the Lipschitz constant $\|a\|_1$. It's significance would be emphasized shortly.

Theorem 5.1. *Let $a \in \mathbb{R}^n$ satisfy $\sum_{i=1}^n a_i = 1$. The vector $c = (\frac{1}{n}, \dots, \frac{1}{n})$ minimizes the Lipschitz constant $\|a\|_q$.*

Proof: We use the definition in Definition 1.1. Then,

$$\|a\|_q = (\sum_{i=1}^n a_i^q)^{\frac{1}{q}}$$

Since $t \mapsto t^{\frac{1}{q}}$ is monotonically increasing assuming $p \in [1, \infty]$, we could reduce the minimization from minimizing $(\sum_{i=1}^n a_i^q)^{\frac{1}{q}}$ to minimizing $\sum_{i=1}^n a_i^q$.

We now use the Lagrange multiplier to find the minimizer of $f(a) = \sum_{i=1}^n a_i^q$ with the constraint $g(a) = \sum_{i=1}^n a_i - 1$. We will solve the system $\nabla f(a) = \lambda \nabla g(a)$, where λ is the Lagrange multiplier.

Solving this system, we get that $(qa_1^{q-1}, \dots, qa_n^{q-1}) = \lambda(1, \dots, 1)$, so $\forall i \in \{1, \dots, n\}, qa_i^{q-1} = \lambda$. This would mean that $\forall i, j \in \{1, \dots, n\}, i \neq j, a_i = a_j$. So, all of the components of a are indeed equal.

Since we assumed that $\sum_{i=1}^n a_i = 1$, if we set $a_i = c_i$ for some $c_i \in \mathbb{R}$, then $nc = 1 \implies c = \frac{1}{n}$. That is, $c = (c_1, \dots, c_n) = (\frac{1}{n}, \dots, \frac{1}{n})$.

In particular, notice $\forall x \in \mathbb{R}^n, \langle c, x \rangle = \frac{1}{n}(\sum_{i=1}^n x_i) = \hat{\mu}$. □

One might question that whether or not the sample mean is the unique minimizer of the Lipschitz constant $\|a\|_q$. We will prove here that it is. While the proofs in this section would not be given for clarity since it uses convex optimization (For more about convex optimization, see [2]), some sources of the proofs are given for the further readings.

We first introduce the notions of a convex function and a convex domain.

Definition 5.2 (Convex function). *A function $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$ is convex if for all $t \in [0, 1]$ and $x_1, x_2 \in X$,*

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2)$$

If we could restrict it to the case where for all $t \in [0, 1]$ and $x_1, x_2 \in X$,

$$f(tx_1 + (1 - t)x_2) < tf(x_1) + (1 - t)f(x_2)$$

we call f a strictly convex function.

Definition 5.3 (Concave function). *A function $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$ is convex if for all $t \in [0,1]$ and $x_1, x_2 \in X$,*

$$f(tx_1 + (1-t)x_2) \geq tf(x_1) + (1-t)f(x_2)$$

If we could restrict it to the case where for all $t \in [0,1]$ and $x_1, x_2 \in X$,

$$f(tx_1 + (1-t)x_2) > tf(x_1) + (1-t)f(x_2)$$

we call f a strictly convex function.

Definition 5.4 (Convex domain). *A domain $X \subseteq \mathbb{R}^n$ is convex if $\forall x_1, x_2 \in X$, there exists a line segment between x_1, x_2 in X.*

We shall now show that any local minimum of a strictly convex function in a convex domain is indeed a unique global minimum.

Lemma 5.5. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a strictly convex function on a convex set C. Then, the local minimum of f over C is also the unique global minimum.*

Proof: Omitted for clarity. A proof is outlined in [3]. □

Remark 5.6. While the proof in [3] uses the closure of $\mathbb{R}$ as the codomain, in finite dimensions, $\overline{\mathbb{R}} = \mathbb{R}$.

We now prove that $\|a\|_q$ is indeed a convex function.

Theorem 5.7. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined as $f(a) = \|a\|_q$ for $q \in (1, \infty)$. Then, f is a strictly convex function.*

Proof: First, we will prove that $f(a) = |a|^q$ is strictly convex. Notice that $f''(a) = q(q-1)|a|^{q-2} > 0$ for $q \in (1,\infty)$, so f is strictly convex and so $\sum_{i=1}^{n} |a_i|^q$ is also strictly convex since the sum of convex functions is also convex.

Thus, $\|ta + (1-t)b\|_q^q < t\|a\|_q^q + (1-t)\|b\|_q^q$ for $t \in [0,1]$, since each component is strictly convex. Since $f(t) = t^{1/q}$ is monotonically increasing, we have:

$$\|ta + (1-t)b\|_q = \left(\|ta + (1-t)b\|_q^q\right)^{1/q} < \left(t\|a\|_q^q + (1-t)\|b\|_q^q\right)^{1/q}$$

Then, notice that $g(t) = t^{1/q}$ is a concave function, so:

$$\left(t\|a\|_q^q + (1-t)\|b\|_q^q\right)^{1/q} \leq t(\|a\|_q) + (1-t)(\|b\|_q)$$

Thus,

$$\|ta + (1-t)b\|_q < \left(t\|a\|_q^q + (1-t)\|b\|_q^q\right)^{1/q} \leq t(\|a\|_q) + (1-t)(\|b\|_q)$$

So $f(a) = \|a\|_q$ is a strictly convex function. □

Now, we build up to the final theorem. That is, the sample mean is the **unique** minimizer of the Lipschitz constant $\|a\|_q$.

Theorem 5.8. *Let $a \in \mathbb{R}^n$ be any vector such that $\sum_{i=1}^{n} a_i = 1$. let $x \in \mathbb{R}^n$. Then, the vector $c = (\frac{1}{n}, \ldots, \frac{1}{n})$ uniquely minimizes the Lipschitz constant $\|a\|_q$, where $\frac{1}{p} + \frac{1}{q} = 1$, and the associate linear estimator $\langle c, x\rangle$ for all $x \in \mathbb{R}^n$ is the sample mean $\hat{\mu}$.*

Proof: First, by Theorem 5.7, $f(a) = \|a\|_q$ is a strictly convex function. Also, by Theorem 5.1, *the minimum of $f(a) = \|a\|_q$ with restricted to $\sum_{i=1}^n a_i = 1$ is $c = (\frac{1}{n}, \ldots, \frac{1}{n})$. We will now prove that this is the unique minimum. Since the affine hyperplane $\sum_{i=1}^n a_i = 1$ in $\mathbb{R}^n$ is a convex domain, by Lemma 5.5, the minimum $c = (\frac{1}{n}, \ldots, \frac{1}{n})$ is indeed the global unique minimum of $\|a\|_q$. Also, by the definition of the dot product, $\forall x \in \mathbb{R}^n, \langle c, x\rangle = \frac{1}{n}(\sum_{i=1}^n x_i) = \hat{\mu}$.* $\square$

Remark 5.9. Another way to view Theorem 5.8 is to prove it as a direct corollary of Jensen's inequality.

6 Conclusion and Further Directions

We have shown that all linear estimators (or linear functionals in $\mathbb{R}^n$) are Lipschitz continuous, which gives us a sense of how well, or how robust, the estimator is against error in data inputs (data perturbations). We have also shown that the vector $c = (\frac{1}{n}, \ldots, \frac{1}{n})$, the precise vector in $\langle c, x\rangle$ for some $x \in \mathbb{R}^n$, which is the sample mean, is the unique minimizer of the Lipschitz constant of any linear estimator. A natural direction for further work is to explore these ideas in L^p function spaces, or to extend such notions to infinite dimensions like in Banach spaces. Another further direction would be to investigate the behaviour of non-linear estimators, such as M-estimators, and compare their behaviors with the linear estimators in this paper.

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